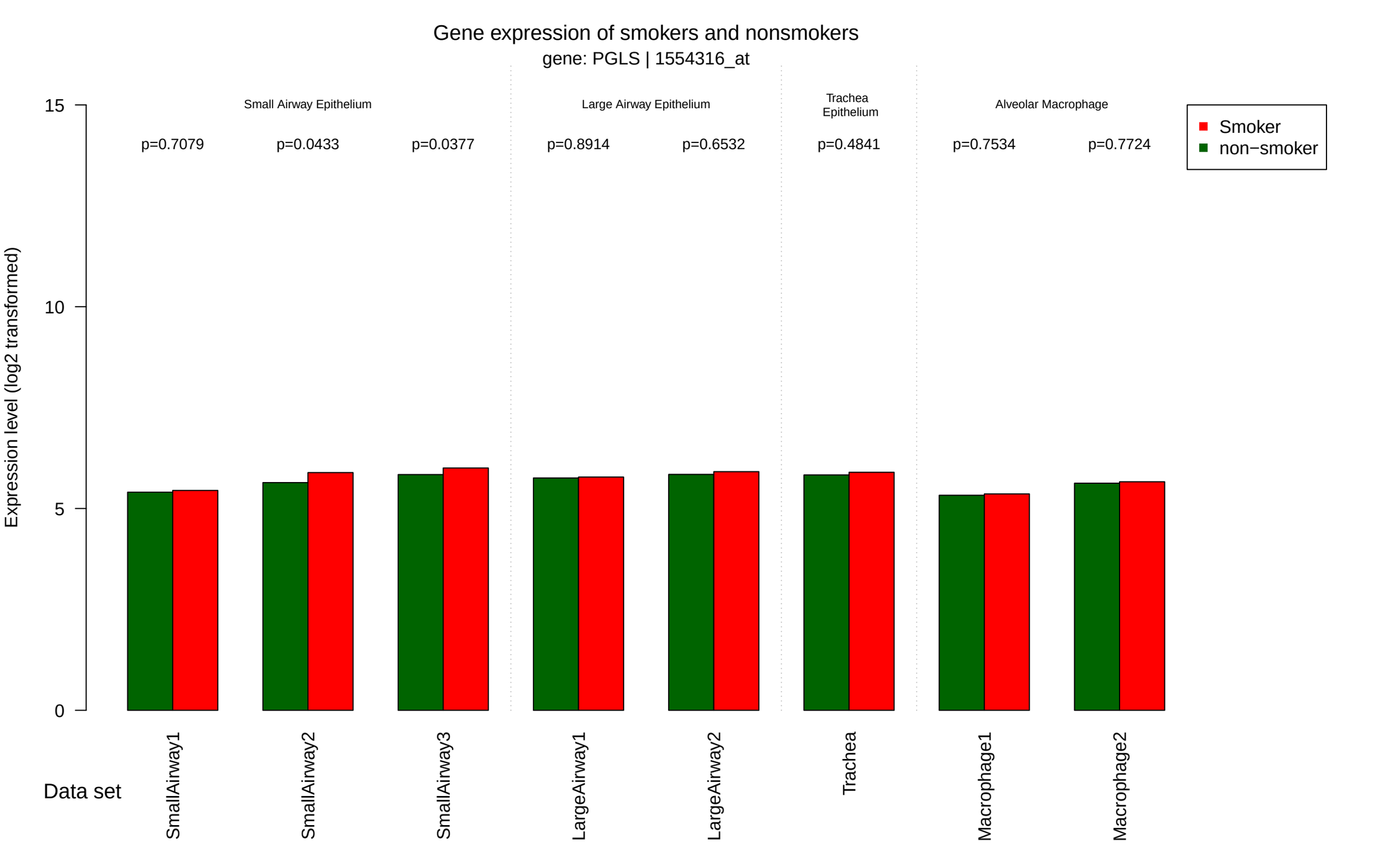
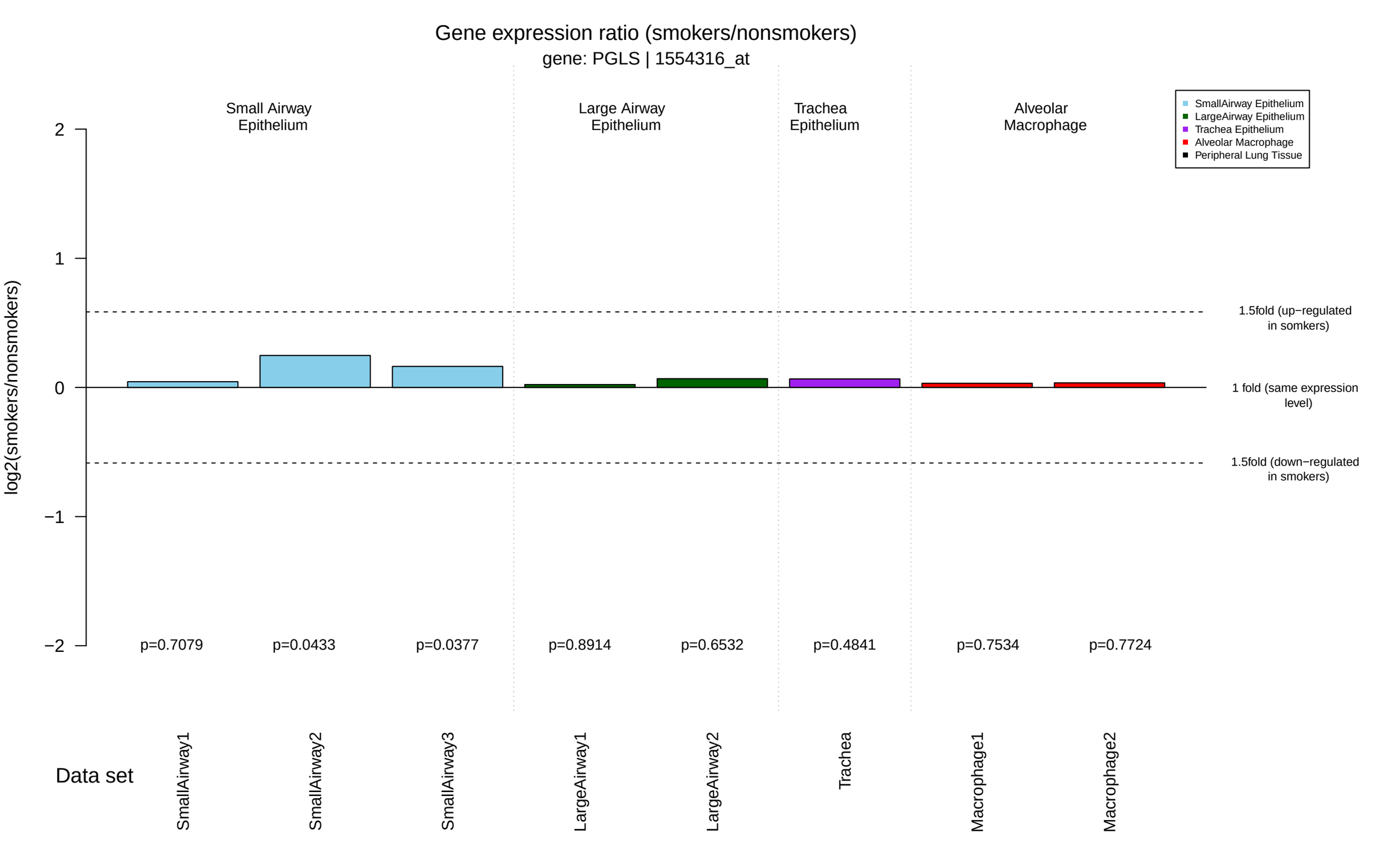
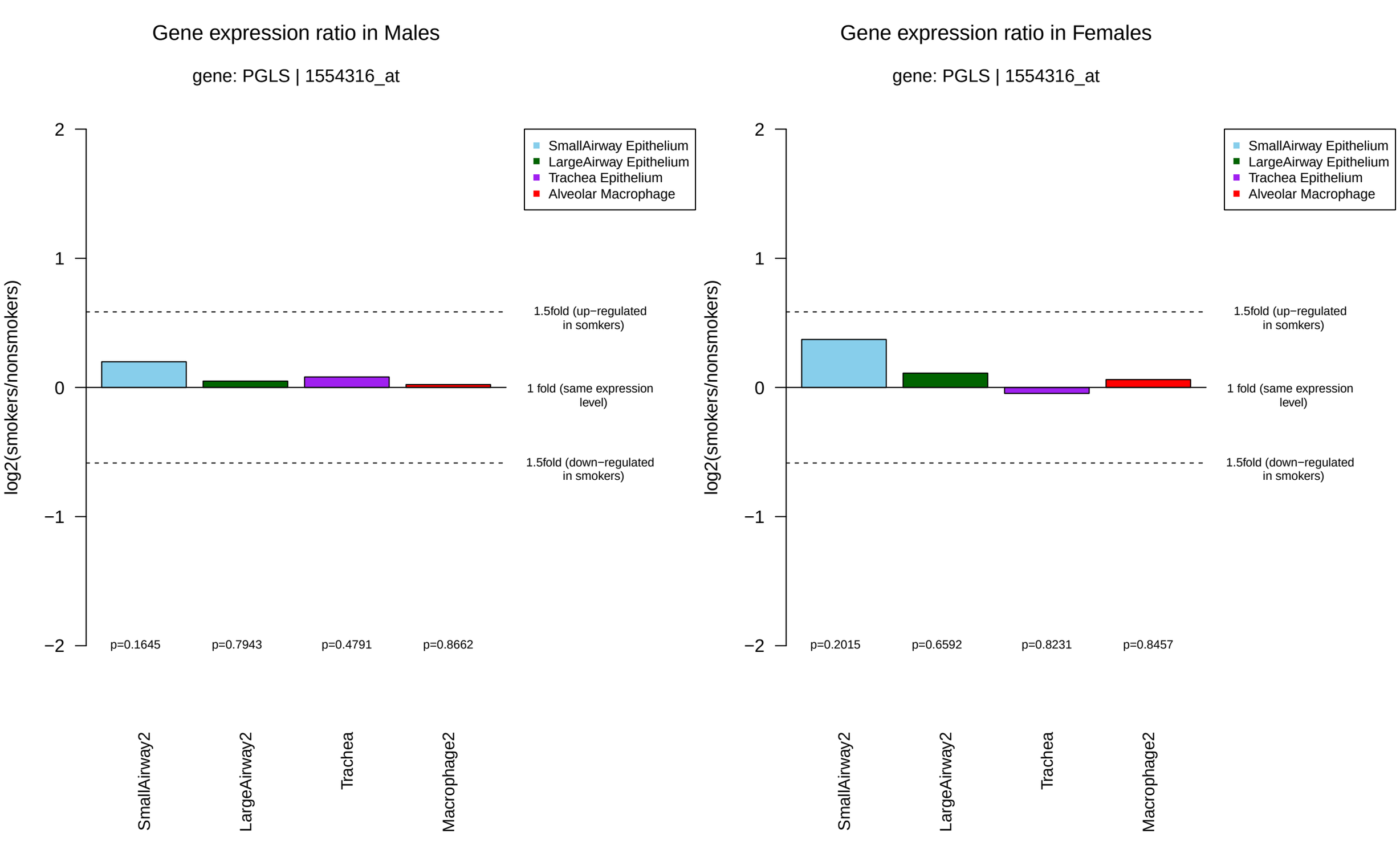
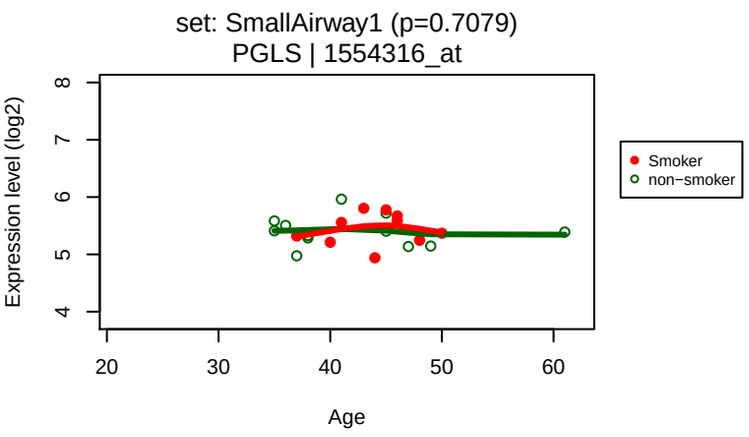
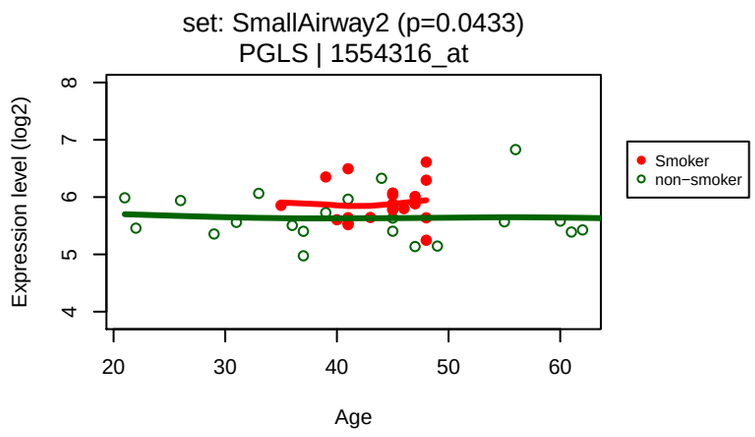
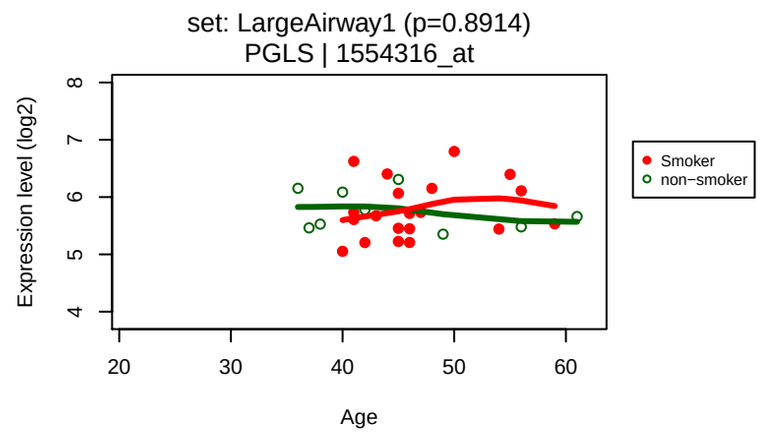

Gene expression of smokers and nonsmokers
gene: PGLS | 1554316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7079
p=0.0433
p=0.0377
p=0.8914
p=0.6532
p=0.4841
p=0.7534
p=0.7724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

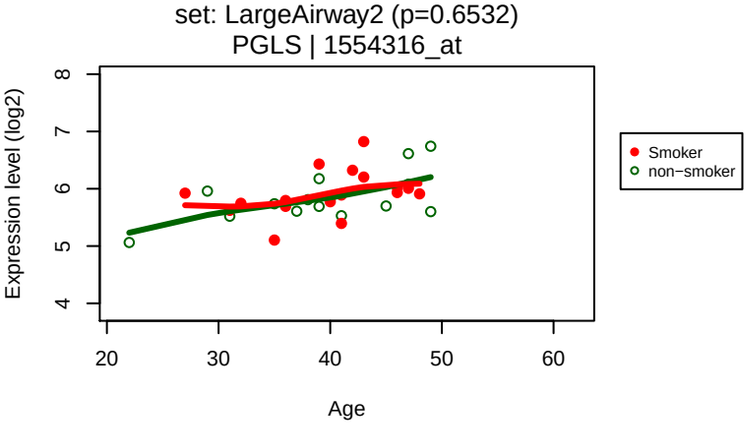

Gene expression ratio (smokers/nonsmokers)
gene: PGLS | 1554316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7079
p=0.0433
p=0.0377
p=0.8914
p=0.6532
p=0.4841
p=0.7534
p=0.7724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

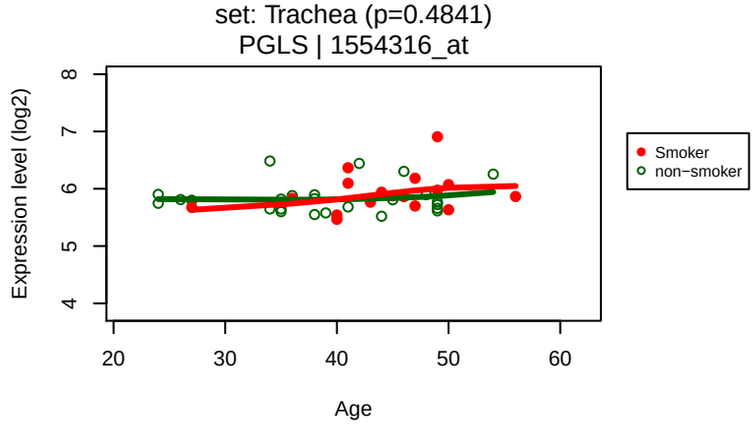

Gene expression ratio in Males
gene: PGLS | 1554316_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1645
p=0.7943
p=0.4791
p=0.8662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PGLS | 1554316_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2015
p=0.6592
p=0.8231
p=0.8457
SmallAirway2
LargeAirway2
Trachea
Macrophage2

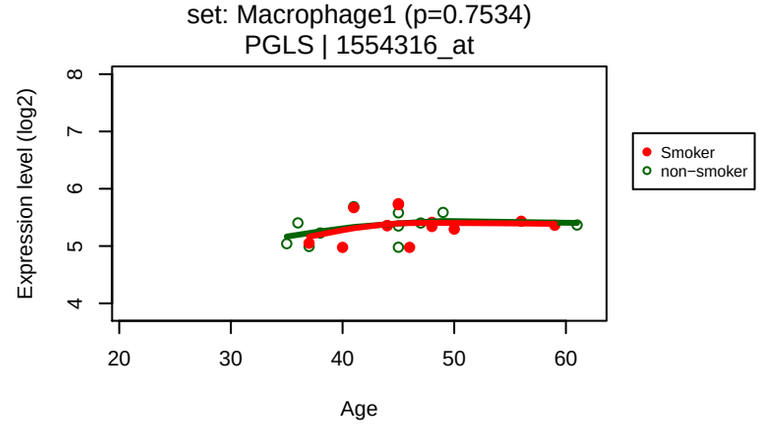

set: SmallAirway1 (p=0.7079)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker

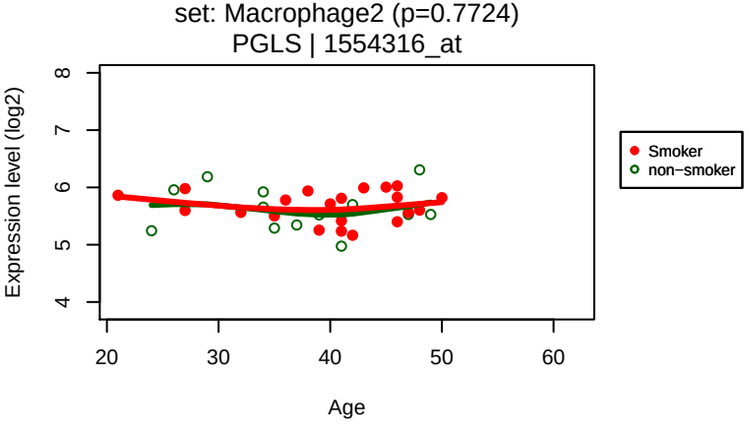

set: SmallAirway2 (p=0.0433)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker

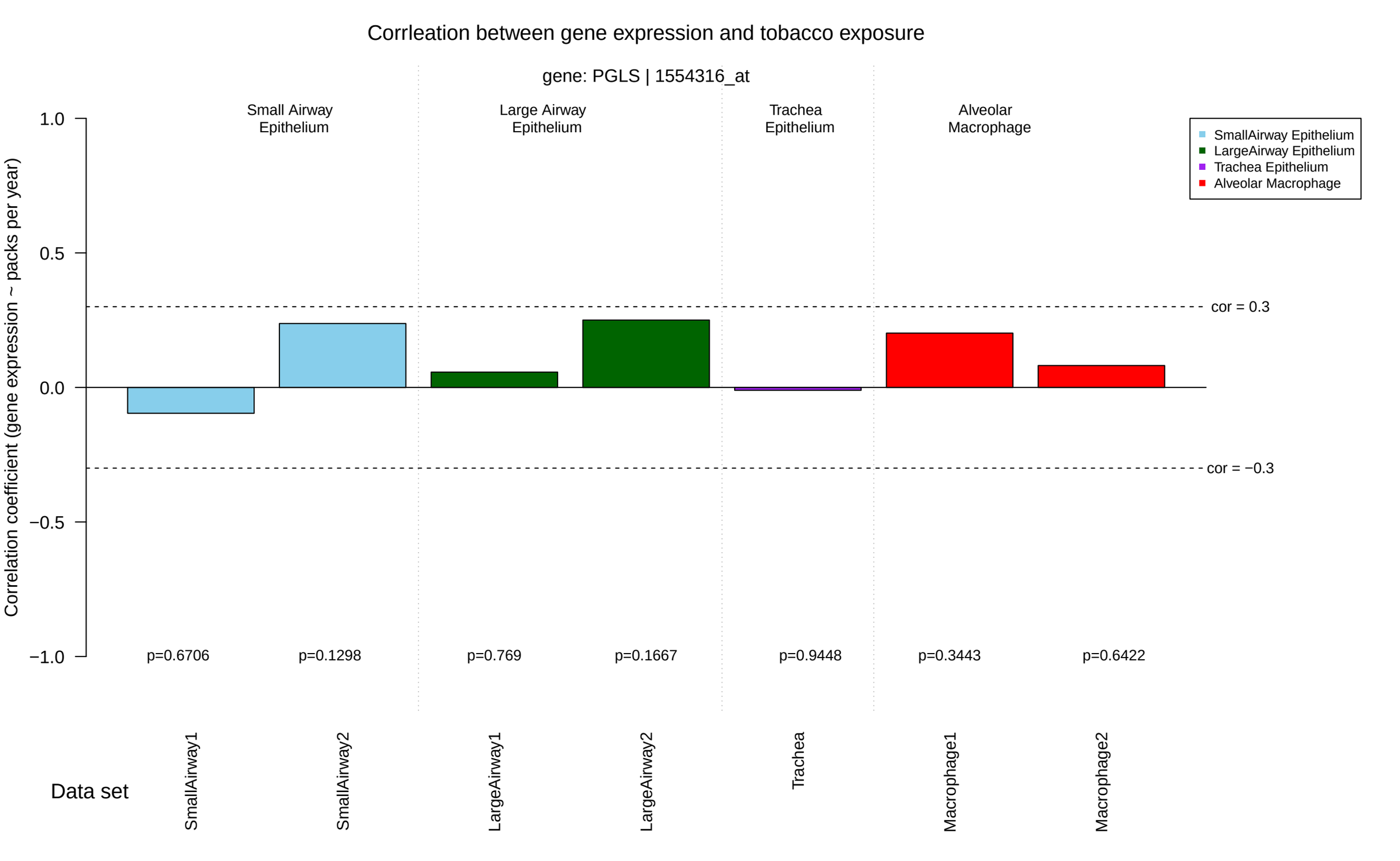

set: LargeAirway1 (p=0.8914)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6532)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4841)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7534)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7724)
PGLS | 1554316_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PGLS | 1554316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6706
p=0.1298
p=0.769
p=0.1667
p=0.9448
p=0.3443
p=0.6422
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2